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PRS821 PROGRAMMING FOR SECURITY PERSONNEL, SEM 1, 2026
WEEK 12 BLOCKCHAIN, MULTICHAIN AND EXPLOIT DEPLOYMENT

Blockchain

At the heart of bitcoin lies the blockchain, a global decentralized ledger which stores the full history of all bitcoin transactions. The blockchain is verified and stored by every node in the bitcoin network, of which there are approximately 6,000 in June 2012. The bitcoin protocol ensures that, barring temporary discrepancies, every node in the network has the same version of the blockchain, without requiring this consensus to be determined by a central authority. Another key feature of bitcoin is that nodes can join or leave the network at any time, without disrupting the functioning of other nodes or the ongoing processing of transactions. New transactions can be created by any node and are propagated across the network in a peer to peer fashion. Any node can take a set of these pending transactions and create ("mine") a new block containing them together with a link to the previous block. The new block "confirms" the transactions and is also propagated across the network. To prevent minority control over mining, bitcoin uses "proof of work" to make it computationally difficult and expensive to create a new block.

MultiChain

MultiChain is an off-the-shelf platform for the creation and deployment of private blockchains, either within or between organizations. It aims to overcome a key obstacle to the deployment of blockchain technology in the institutional financial sector, by providing the privacy and control required in an easy to use package. Like the Bitcoin Core software from which it is derived, MultiChain supports Windows, Linux and Mac servers and provides a simple API and command line interface.

Exploit Deployment

An exploit (in its noun form) is a segment of code or a program that maliciously takes advantage of vulnerabilities or security flaws in software or hardware to infiltrate and initiate a denial-of-service (DoS) attack or install malware, such as spyware, ransomware, Trojan horses, worms, or viruses. So the exploit is not the malware itself but is used to deliver the malware. To exploit (in its verb form) is to successfully carry out such an attack.

Exploits are often the first part of a larger attack. Hackers scan for outdated systems that contain critical vulnerabilities, which they then exploit by deploying targeted malware. Exploits often include shellcode, which is a small malware payload used to download additional malware from attacker-controlled networks. Shellcode allows hackers to infect devices and infiltrate organizations.

Exploit kits are more comprehensive tools that contain a collection of exploits. These kits scan devices for different kinds of software vulnerabilities and, if any are detected, deploy additional malware to further infect a device. Kits can use exploits targeting various software, including Adobe Flash Player, Adobe Reader, Internet Explorer, Oracle Java, and Sun Java.

The most common method used by attackers to distribute exploits and exploit kits is through webpages, but exploits can also arrive in emails. Some websites unknowingly and unwillingly host malicious code and exploits in their ads.

How does an exploit work?

For exploits to be effective, many vulnerabilities require an attacker to initiate a series of suspicious operations to set up an exploit. Typically, a majority of the vulnerabilities are result of a software or system architecture bug. Attackers write their code to take advantage of these vulnerabilities and inject various types of malware into the system.

When developers produce an operating system (OS) for a device, write code for software, or develop an application, bugs often appear due to inherent imperfections. These bugs can create a vulnerability in the system, and an exploit searches out such vulnerabilities and looks for a way to exploit databases and networks or systems.

If the bug is not reported and “patched,” it becomes an entryway, so to speak, for cyber criminals to conduct an exploit. With so many devices connected together in the modern world, as in the Internet of Things (IoT), for example, an exploit does not just compromise a singular device, but it can become a security vulnerability for a whole network.

How do I defend against exploits?

Many software vendors patch known bugs to remove the vulnerability. Security software also helps by detecting, reporting, and blocking suspicious operations. It prevents exploits from occurring and damaging computer systems, regardless of what malware the exploit was trying to initiate.